Q1. JEE Main 2026 (21 January Shift 2)

Let $\mathrm{A}=\left\{x:\left|x^2-10\right| \leq 6\right\}$ and $\mathrm{B}=\{x:|x-2|>1\}$. Then

(1) $\mathrm{A} \cup \mathrm{B}=(-\infty, 1] \cup(2, \infty)$

(2) $\mathrm{B}-\mathrm{A}=(-\infty,-4) \cup(-2,1) \cup(4, \infty)$

(3) $\mathrm{A}-\mathrm{B}=[2,3)$

(4) $\mathrm{A} \cap \mathrm{B}=[-4,-2] \cup[3,4]$

Q2. JEE Main 2026 (21 January Shift 2)

Let $\mathrm{A}=\{2,3,5,7,9\}$. Let R be the relation on A defined by $x \mathrm{R} y$ if and only if $2x \leq 3y$. Let l be the number of elements in R, and m be the minimum number of elements required to be added in R to make it a symmetric relation. Then $l+\mathrm{m}$ is equal to :

(1) 21 (2) 27 (3) 23 (4) 25

Q3. JEE Main 2026 (23 January Shift 1)

Let $\mathrm{A}=\{-2,-1,0,1,2,3,4\}$. Let R be a relation on A defined by $x \mathrm{R} y$ if and only if $2x+y \leqslant 2$. Let l be the number of elements in R. Let m and n be the minimum number of elements required to be added in R to make it reflexive and symmetric relations respectively. Then $l+m+n$ is equal to :

(1) 35 (2) 34 (3) 33 (4) 32

Q4. JEE Main 2026 (22 January Shift 2)

The number of elements in the relation $\mathrm{R}=\left\{(x, y): 4x^2+y^2<52, x, y \in \mathbf{Z}\right\}$ is

(1) 77 (2) 67 (3) 86 (4) 89

Q5. JEE Main 2026 (22 January Shift 2)

Let S be the set of the first 11 natural numbers. Then the number of elements in $A=\{B \subseteq S: n(B) \geqslant 2$ and the product of all elements of B is even $\}$ is ____ .

Q6. JEE Main 2026 (22 January Shift 1)

Let the relation R on the set $\mathrm{M}=\{1,2,3, \ldots, 16\}$ be given by $\mathrm{R}=\{(x, y): 4y=5x-3, x, y \in \mathrm{M}\}$. Then the minimum number of elements required to be added in R, in order to make the relation symmetric, is equal to

(1) 4 (2) 1 (3) 2 (4) 3

Q7. JEE Main 2026 (23 January Shift 2)

Let $\mathrm{A} = \{0, 1, 2, \ldots, 9\}$. Let R be a relation on A defined by $(x, y) \in \mathrm{R}$ if and only if $|x - y|$ is a multiple of 3.
Given below are two statements:
Statement I: $n(\mathrm{R}) = 36$.
Statement II: R is an equivalence relation.
In the light of the above statements, choose the correct answer from the options given below
(1) Statement I is incorrect but Statement II is correct
(2) Statement I is correct but Statement II is incorrect
(3) Both Statement I and Statement II are incorrect
(4) Both Statement I and Statement II are correct

Q8. JEE Main 2026 (21 January Shift 1)

The number of relations, defined on the set $\{\mathrm{a}, \mathrm{b}, \mathrm{c}, \mathrm{d}\}$, which are both reflexive and symmetric, is equal to:
(1) 16 (2) 1024 (3) 64 (4) 256

Q9. JEE Main 2026 (24 January Shift 1)

Let R be a relation defined on the set $\{1, 2, 3, 4\} \times \{1, 2, 3, 4\}$ by $\mathrm{R} = \{((a, b), (c, d)) : 2a + 3b = 3c + 4d\}$. Then the number of elements in R is
(1) 18 (2) 6 (3) 15 (4) 12

ANSWERS AND SOLUTIONS

1. (2) **2.** (4) **3.** (3) **4.** (1) **5.** 1979 **6.** (3) **7.** (1) **8.** (3)
9. (4)

1. **(2)** $A = \{x : |x^2 - 10| \leq 6\}$: $-6 \leq x^2 - 10 \leq 6 \Rightarrow 4 \leq x^2 \leq 16$, so $A = [-4, -2] \cup [2, 4]$.
$B = \{x : |x - 2| > 1\} = (-\infty, 1) \cup (3, \infty)$.
$A^c = (-\infty, -4) \cup (-2, 2) \cup (4, \infty)$.
$B - A = B \cap A^c = [(-\infty, 1) \cap A^c] \cup [(3, \infty) \cap A^c]$
$= [(-\infty, -4) \cup (-2, 1)] \cup [(4, \infty)]$
$= (-\infty, -4) \cup (-2, 1) \cup (4, \infty)$.

2. **(4)** Given $A = \{2, 3, 5, 7, 9\}$ and xRy iff $2x \leq 3y$, i.e., $y \geq \frac{2x}{3}$.
For $x = 2$: $y \in \{2, 3, 5, 7, 9\} \to 5$ pairs
For $x = 3$: $y \in \{2, 3, 5, 7, 9\} \to 5$ pairs
For $x = 5$: $y \in \{5, 7, 9\} \to 3$ pairs
For $x = 7$: $y \in \{5, 7, 9\} \to 3$ pairs
For $x = 9$: $y \in \{7, 9\} \to 2$ pairs
So $l = 18$.
For symmetry, pairs in R whose reversal is missing: $(2, 5), (2, 7), (2, 9), (3, 5), (3, 7), (3, 9), (5, 9)$ require adding $(5, 2), (7, 2), (9, 2), (5, 3), (7, 3), (9, 3), (9, 5) \to m = 7$.
$l + m = 18 + 7 = 25$.

3. **(3)** Given $A = \{-2, -1, 0, 1, 2, 3, 4\}$ and xRy iff $2x + y \leqslant 2$.
For $x = -2$: $y \leqslant 6$, all 7 elements valid (7 pairs).
For $x = -1$: $y \leqslant 4$, all 7 elements valid (7 pairs).
For $x = 0$: $y \leqslant 2 \Rightarrow y \in \{-2, -1, 0, 1, 2\}$ (5 pairs).
For $x = 1$: $y \leqslant 0 \Rightarrow y \in \{-2, -1, 0\}$ (3 pairs).
For $x = 2$: $y \leqslant -2 \Rightarrow y \in \{-2\}$ (1 pair).
For $x = 3, 4$: no valid y.
So $l = 7 + 7 + 5 + 3 + 1 = 23$.
For reflexive: $(x, x) \in R \Rightarrow 3x \leqslant 2 \Rightarrow x \leqslant 2/3$. Already present for $x = -2, -1, 0$. Missing for $x = 1, 2, 3, 4$, so $m = 4$.
For symmetric: need $(y, x) \in R$ for every $(x, y) \in R$. Missing pairs:
$(3, -2), (4, -2), (2, -1), (3, -1), (4, -1), (2, 0)$, so $n = 6$.
$l + m + n = 23 + 4 + 6 = 33$.

4. **(1)** We need $4x^2 + y^2 < 52$ with $x, y \in \mathbb{Z}$. This requires $x^2 < 13$, so $x \in \{-3, -2, -1, 0, 1, 2, 3\}$.
$x = 0$: $y^2 < 52 \Rightarrow y \in \{-7, \ldots, 7\} \to 15$ values

$x = \pm 1$: $y^2 < 48 \Rightarrow |y| \leq 6 \rightarrow 13$ each $\rightarrow 26$ values

$x = \pm 2$: $y^2 < 36 \Rightarrow |y| \leq 5 \rightarrow 11$ each $\rightarrow 22$ values

$x = \pm 3$: $y^2 < 16 \Rightarrow |y| \leq 3 \rightarrow 7$ each $\rightarrow 14$ values

Total $= 15 + 26 + 22 + 14 = 77$

5. **(1979)** The set $S = \{1, 2, 3, \ldots, 11\}$ has 11 elements.

We need to count subsets B with $n(B) \geq 2$ and even product.

A product is even iff at least one element is even.

Equivalently, count complements: a product is odd iff all elements are odd.

The odd numbers in S are $\{1, 3, 5, 7, 9, 11\}$ (6 elements).

Subsets of odd numbers: $2^6 = 64$.

Among these, those with $n(B) \geq 2$ are all except empty set (1) and singletons (6): $64 - 7 = 57$.

Total subsets with $n(B) \geq 2$: $2^{11} - 1 - 11 = 2036$.

Subsets with even product and $n(B) \geq 2$: $2036 - 57 = 1979$.

6. **(3)** Given $R = \{(x, y) : 4y = 5x - 3,\ x, y \in M\}$ where $M = \{1, 2, \ldots, 16\}$.

From $4y = 5x - 3$, we get $y = \frac{5x-3}{4}$.

For y to be a positive integer, $5x - 3 \equiv 0 \pmod 4$, i.e., $x \equiv 3 \pmod 4$.

So $x = 3, 7, 11, 15$.

$x = 3 \Rightarrow y = 3$,

$x = 7 \Rightarrow y = 8$,

$x = 11 \Rightarrow y = 13$,

$x = 15 \Rightarrow y = 18 \notin M$.

$R = \{(3, 3),\ (7, 8),\ (11, 13)\}$.

For symmetry, $(7, 8) \in R$ requires $(8, 7) \in R$ and $(11, 13) \in R$ requires $(13, 11) \in R$.

$(3, 3)$ is already symmetric.

Minimum elements to be added $= 2$.

7. **(1)** The relation R on $A = \{0, 1, 2, \ldots, 9\}$ is defined by $(x, y) \in R$ iff $|x - y|$ is a multiple of 3, i.e.,

$x \equiv y \pmod 3$.

Statement II:

R is reflexive ($|x - x| = 0$, a multiple of 3), symmetric ($|x - y| = |y - x|$), and transitive (if $x \equiv y$ and $y \equiv z \pmod 3$, then $x \equiv z \pmod 3$). So R is an equivalence relation. Statement II is correct.

Statement I:

The equivalence classes are $\{0, 3, 6, 9\}$ (4 elements), $\{1, 4, 7\}$ (3 elements), $\{2, 5, 8\}$ (3 elements).

$n(R) = 4^2 + 3^2 + 3^2 = 16 + 9 + 9 = 34 \neq 36$. Statement I is incorrect.

8. **(3)** For a set with $n = 4$ elements, reflexive condition requires all diagonal pairs $(a, a), (b, b), (c, c), (d, d)$ to be included.

For symmetric relations, non-diagonal pairs come in symmetric pairs. Number of such pairs = $\binom{4}{2} = 6$.

For each pair, we either include both elements or exclude both $\rightarrow$ 2 choices each.

Total reflexive and symmetric relations = $2^6 = 64$.

9. **(4)** We need $2a + 3b = 3c + 4d$ where $a, b, c, d \in \{1, 2, 3, 4\}$.

Possible values of $2a + 3b$: range from 5 to 20.

Possible values of $3c + 4d$: range from 7 to 28.

For each common value k, the number of elements in R equals (number of (a, b) with $2a + 3b = k$) $\times$ (number of (c, d) with $3c + 4d = k$).

$k = 7$: $(a, b) = (2, 1)$, $(c, d) = (1, 1) \Rightarrow 1 \times 1 = 1$

$k = 10$: $(2, 2)$, $(2, 1) \Rightarrow 1$

$k = 11$: $(1, 3)$, $(4, 1)$, $(1, 2) \Rightarrow 2$

$k = 13$: $(2, 3)$, $(3, 1) \Rightarrow 1$

$k = 14$: $(1, 4)$, $(4, 2)$, $(2, 2) \Rightarrow 2$

$k = 15$: $(3, 3)$, $(1, 3) \Rightarrow 1$

$k = 16$: $(2, 4)$, $(4, 1) \Rightarrow 1$

$k = 17$: $(4, 3)$, $(3, 2) \Rightarrow 1$

$k = 18$: $(3, 4)$, $(2, 3) \Rightarrow 1$

$k = 20$: $(4, 4)$, $(4, 2) \Rightarrow 1$

Total $= 1 + 1 + 2 + 1 + 2 + 1 + 1 + 1 + 1 + 1 = 12$